

Which of the following four Fixed-Point Iterations converge to $\sqrt[3]{2}$? Rank the ones that converge from fastest to slowest.

$$(a) \quad x \longrightarrow \frac{2}{3}x + \frac{2}{3x^2}; \quad (b) \quad x \longrightarrow \frac{1}{2}x + \frac{1}{x^2};$$

$$(c) \quad x \longrightarrow \frac{3}{4}x + \frac{1}{2x^2}; \quad (d) \quad x \longrightarrow \frac{2}{3}x + \frac{1}{x^2}.$$

Solution.

Let $\alpha = \sqrt[3]{2}$, so that $\alpha^3 = 2$. For a fixed-point iteration $x_{k+1} = g(x_k)$, we first check that α is a fixed point, and then examine $|g'(\alpha)|$.

(a)

$$g(x) = \frac{2}{3}x + \frac{2}{3x^2}.$$

Since

$$g(\alpha) = \frac{2}{3}\alpha + \frac{2}{3\alpha^2} = \frac{2}{3}\alpha + \frac{1}{3}\alpha = \alpha,$$

 α is a fixed point. Moreover,

$$g'(x) = \frac{2}{3} - \frac{4}{3x^3},$$

and hence

$$g'(\alpha) = \frac{2}{3} - \frac{4}{3(2)} = 0.$$

Therefore, (a) converges quadratically to $\sqrt[3]{2}$.

(b)

$$g(x) = \frac{1}{2}x + \frac{1}{x^2}.$$

Since

$$g(\alpha) = \frac{1}{2}\alpha + \frac{1}{\alpha^2} = \frac{1}{2}\alpha + \frac{1}{2}\alpha = \alpha,$$

 α is a fixed point. Moreover,

$$g'(x) = \frac{1}{2} - \frac{2}{x^3},$$

so

$$g'(\alpha) = \frac{1}{2} - \frac{2}{2} = -\frac{1}{2}.$$

Therefore, (b) converges linearly to $\sqrt[3]{2}$ with asymptotic convergence factor $1/2$.

(c)

$$g(x) = \frac{3}{4}x + \frac{1}{2x^2}.$$

Since

$$g(\alpha) = \frac{3}{4}\alpha + \frac{1}{2\alpha^2} = \frac{3}{4}\alpha + \frac{1}{4}\alpha = \alpha,$$

 α is a fixed point. Moreover,

$$g'(x) = \frac{3}{4} - \frac{1}{x^3},$$

so

$$g'(\alpha) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}.$$

Therefore, (c) converges linearly to $\sqrt[3]{2}$ with asymptotic convergence factor $1/4$.

(d)

$$g(x) = \frac{2}{3}x + \frac{1}{x^2}.$$

However,

$$g(\alpha) = \frac{2}{3}\alpha + \frac{1}{\alpha^2} = \frac{2}{3}\alpha + \frac{1}{2}\alpha = \frac{7}{6}\alpha \neq \alpha.$$

Thus $\sqrt[3]{2}$ is not a fixed point of (d), so (d) cannot converge to $\sqrt[3]{2}$.

Hence, from fastest to slowest,

$$\boxed{(a) > (c) > (b),}$$

where (a) is quadratically convergent, while (c) and (b) are linearly convergent with asymptotic convergence factors $1/4$ and $1/2$, respectively. Iteration (d) does not converge to $\sqrt[3]{2}$.